Curriculum vitae

July 2026

Antonio Bilotta, PhD
February 13, 1968, Crotone (Italia)

Associate professor of Solid and Structural Mechanics (CEAR-06/A - Scienza delle costruzioni)

University of Calabria
Via P. Bucci, 39/C, 87036 Rende (CS) Italia
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Personal web page

EDUCATION

1999	PhD in Computational Mechanics, University of Calabria (Italy).
1995	Degree in Civil Engineering, final rank *110/110 cum laude*, University of Calabria (Italy).

ACADEMIC CAREER

2020-2029	National Scientific Qualification as Full Professor in Solid Mechanics.
2017-2026	National Scientific Qualification as Associate Professor in Solid Mechanics.
2023 ——	Associate Professor in Solid Mechanics, University of Calabria (Italy).
2005-2023	Professor Assistant in Solid Mechanics, University of Calabria (Italy).
2001-2005	Research fellow in *Computational strategies for the nonlinear analysis of structures*, contract n. 113, University of Calabria (Italy).
1999-2001	Postdoctoral research fellow, University of Calabria (Italy).

PARTECIPATION IN RESEARCH PROJECTS

2023-2025	Next Generation EU - PNRR, Tech4You project - Technologies for climate change adaptation and quality of life improvement, n. ECS0000009, CUP H23C22000370006.
2023-2025	Prin-PNRR 2022 project n. P2022XPK7W, CUP H53D23008710001, leader Prof. Antonio Formisano. *RESILIENT: Waste REuse for anti-SeIsmic masonry buiLdIngs with energy-efficiENT behaviour.*
2023-2025	Prin 2022 project n. 2022BN3RAC, CUP D53D23003410006, leader Prof. Marco De Tullio. *ABYSS: Accurate simulation of Bio-hYbrid Soft Swimmers.*
2022-2024	DPC_RELUIS2022_2024 Project, CUP H23C22000630001.
2019-2023	Prin 2017 project no. 2017J4EAYB, leader Prof. Fernando Fraternali. *Multiscale innovative materials and structures (MIMS).*
2015	MADAR (POR FESR), project no. J85G09000350002, leader Prof. Margerita Solci. *Mathematical models for the simulation of structural degradation phenomena in archaeological areas*, https://rinnovarelatutela.wordpress.com/madar/.

2013-2016	Prin 2010-11 project no. 2010NRBMTP, leader Prof. Raffaele Casciaro. *Models and algorithms for non-linear analysis of structures and validation of performance-based design rules.*
2008-2010	Prin 2007 project no. 20072JZL8K, leader Prof. Raffaele Casciaro. *Performance-based modeling and analysis of non-linear structures.*
2006-2007	Prin 2005 project no. 2005083412, leader Prof. Cesare Davini. *Modeling and approximation techniques in advanced problems of materials and structure mechanics.*
2004-2005	Prin 2003 project, leader Prof. Antonino Morassi. *Non-destructive methods for identifying and diagnosing materials and structures.*
2004-2005	Prin 2003 project no. 2003082318, leader Prof. Raffaele Casciaro. *Definition of integrated methods for the structural assessment of masonry buildings.*
2002-2005	Academy of Finland project no. 55375, leader Prof. Jukka Tuhkuri. *A thermomechanics based fracture assessment method for structural components.*
1999-2000	Prin 1998 project no. 9808052678, leader Prof. Raffaele Casciaro. *Development of an integrated strategy for the modeling, analysis and assessment of masonry constructions.*
1998-2001	POP 94/99 MECOM project "Misura 4.4 - Ricerca scientifica e tecnologica", leader Prof. Raffaele Casciaro. *Development and application of computational mechanics in structural design in civil and industrial fields.*
1996-2000	Brite Euram project no. BRPR-CT96-0202, leader Bruno Beràl. *Advanced PRImary COmposites Structures - APRICOS.*

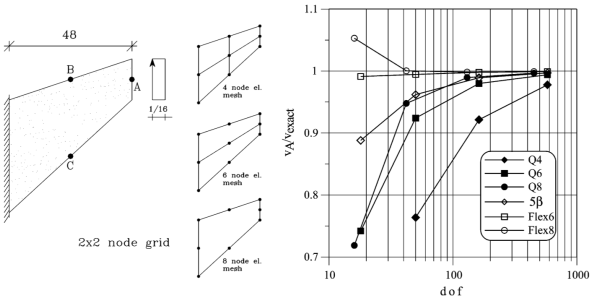
PUBLISHING ACTIVITIES FOR INTERNATIONAL JOURNALS

- Member of the editorial board of the journal *Mathematical Problems in Engineering*, https://www.hindawi.com/journals/mpe/editors/ (starting from July 12, 2016).
- Reviewer for the following journals:
  - *Computers & Structures*
  - *Finite Elements in Analysis & Design*
  - *Meccanica*
  - *International Journal of Solids and Structures*
  - *Computers and Mathematics with Applications*
  - *Journal of Vibration and Control*
  - *Mechanical Systems and Signal Processing*

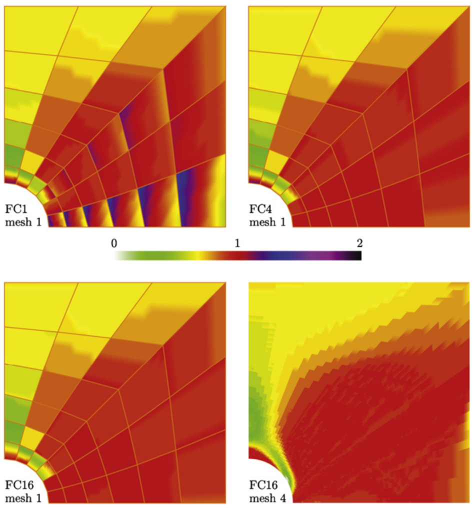
RESEARCH TOPICS

- **Mixed finite element formulations** [4, 8, 20, 22, 33, 37, 41].
- **Other high performance FEM models for structural analysis** [34, 40].
- **Enhanced beam models** [17, 24, 26, 27, 28, 29, 31].
- **Elastoplastic analysis of structures** [17, 19, 20, 22, 32, 33, 37].
- **Models for masonries** [4, 10, 14, 19, 42].
- **Crack growth problems** [40].
- **Inverse problems** [1, 7, 16, 18, 21, 23, 35, 36, 38, 39].
- **Materials** [5, 6].
- **Misc** [2, 3].
- **Applications of Machine Learning** [5, 7, 10].

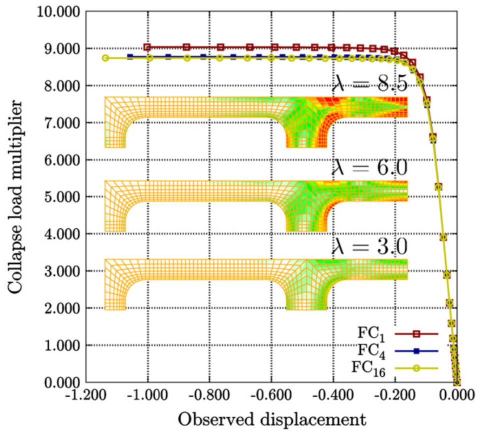
Mixed finite element formulations [4, 8, 20, 22, 33, 37, 41]



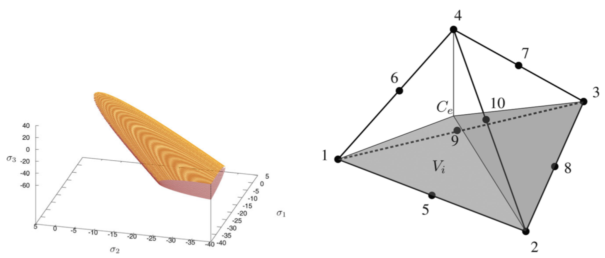
Comp. Meth. Appl. Mech. Engrg. 191 (2002) [41]



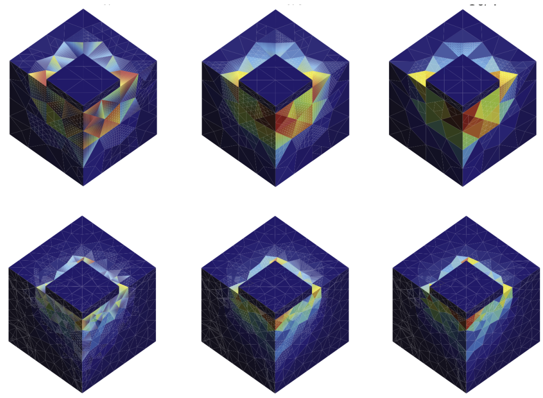
Fin. Elem. An. Des. 47 (2011) [33]



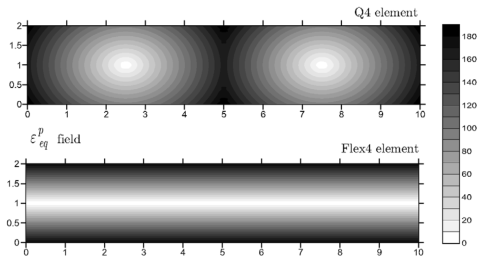
Fin. Elem. An. Des. 47 (2011) [33]



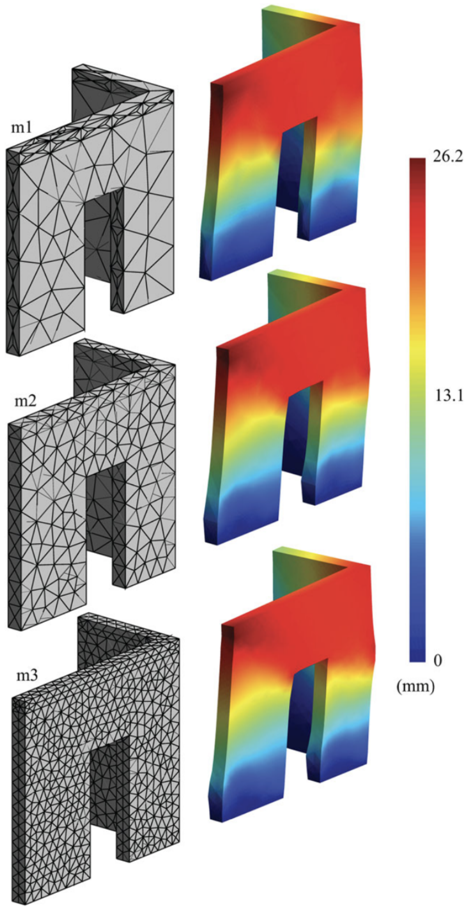
ZAMM 97 (2017) [20]



Fin. Elem. An. Des. 113 (2016) [22]

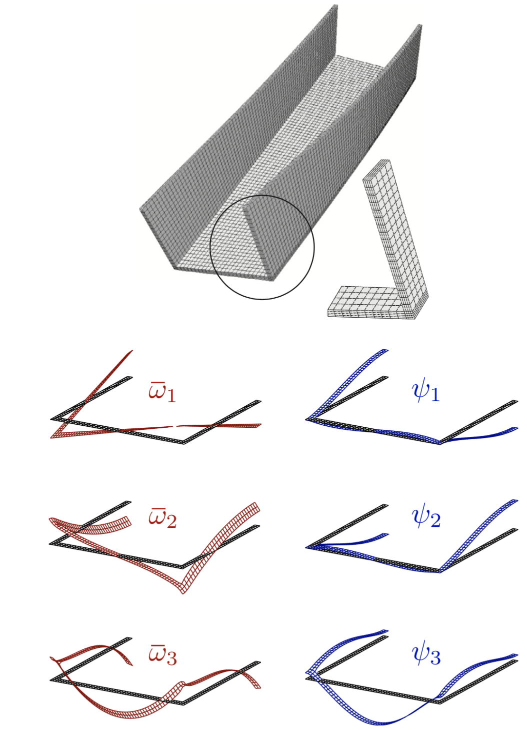


Comp. Meth. Appl. Mech. Engrg. 196 (2007) [37]

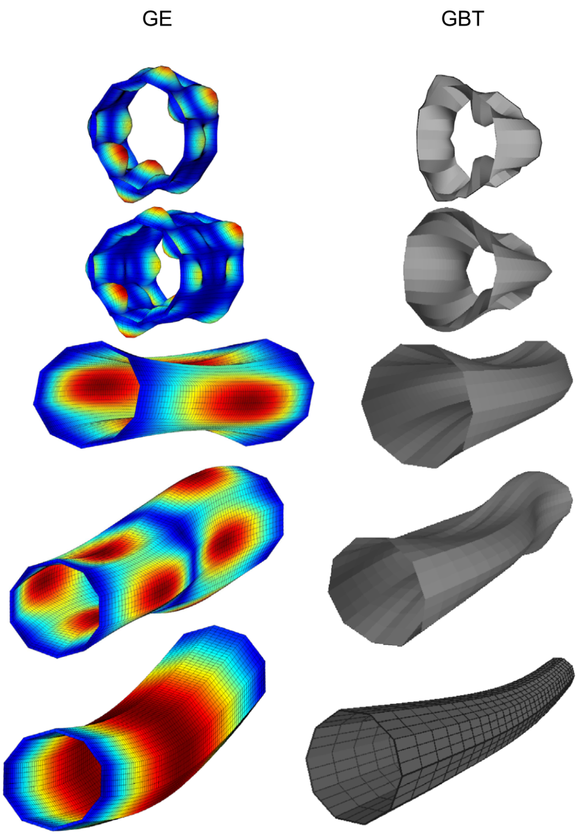


ZAMM 97 (2017) [20]

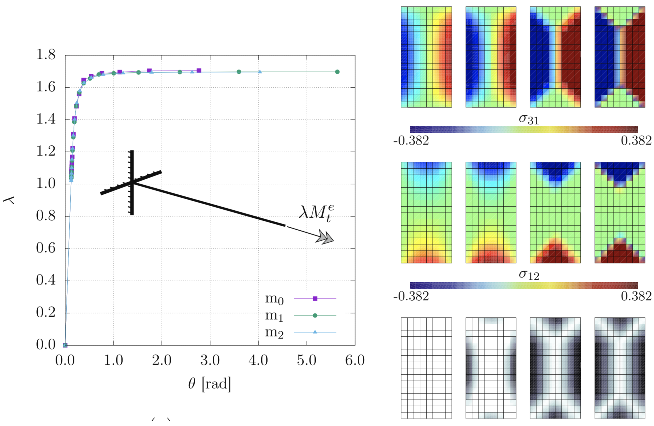
Enhanced beam models [17, 24, 26, 27, 28, 29, 31]



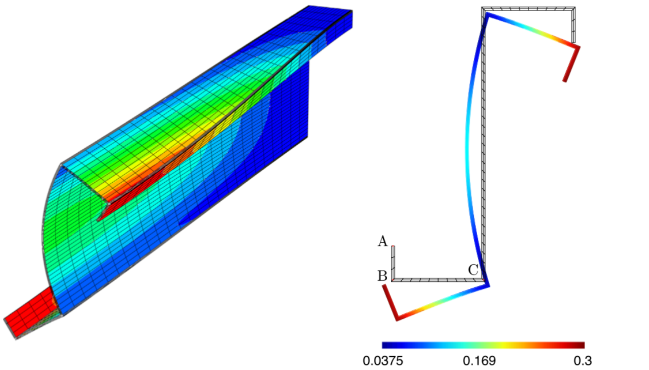
Computer & Structures 121 (2013) [31]



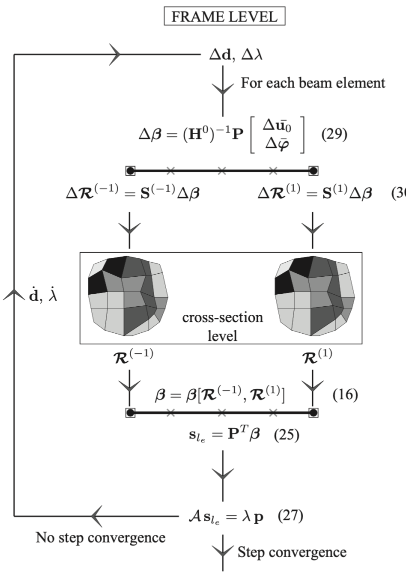
Thin-Walled Structures 100 (2016) [24]



Composite Structures 209 (2019) [17]

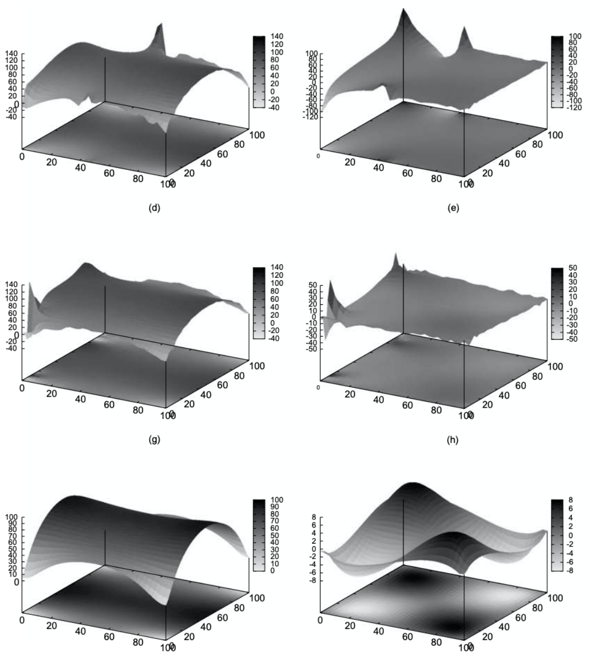


Thin-Walled Structures 74 (2014) [26]

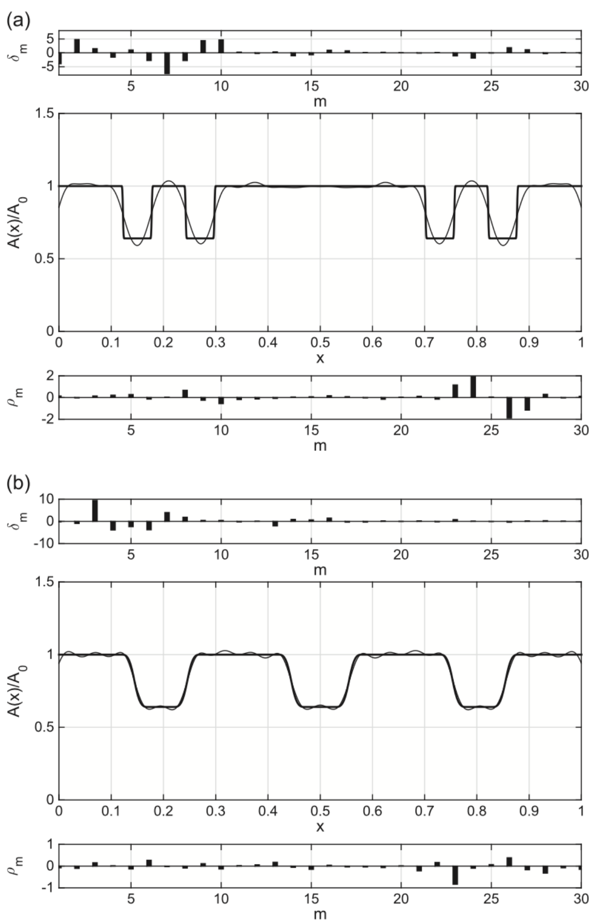


Composite Structures 209 (2019) [17]

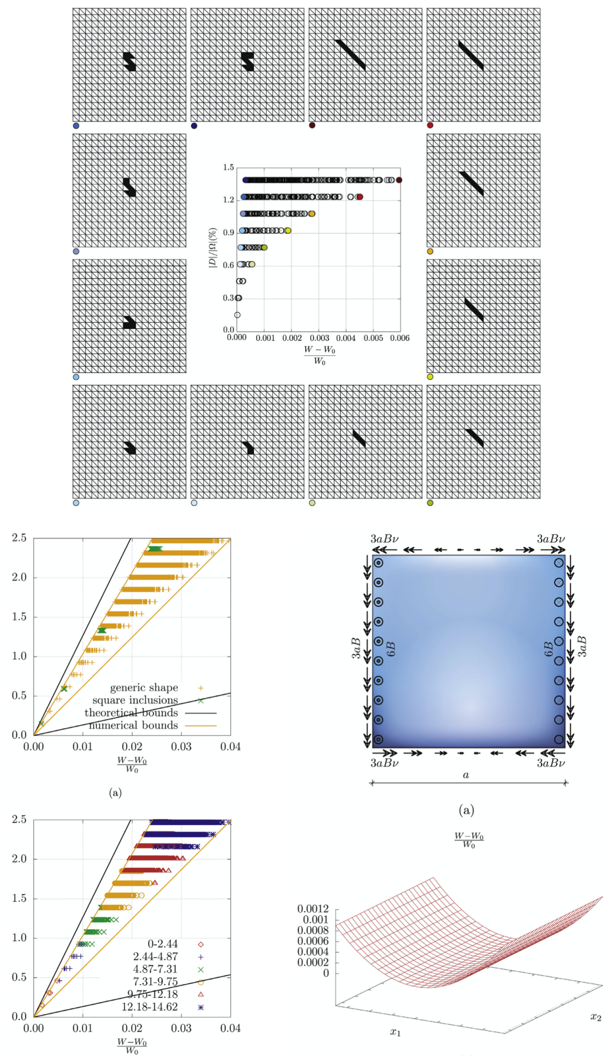
Inverse problems [7, 16, 18, 21, 23, 35, 36, 38, 39]



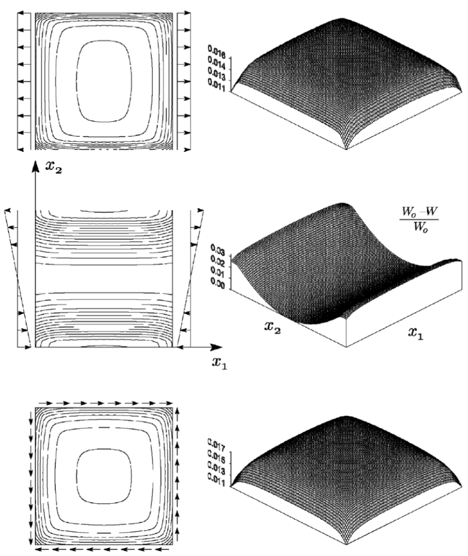
Int. J. Solids and Struct. 46 (2009) [35]



Journal of Sound and Vibration 366 (2016) [23]



Int. J. Solids and Struct. 168 (2019) [16]



J. Comput. Appl. Math. 198 (2007) [38]

TEACHING

2022-present Lecturer of the course in *Solid Mechanics* for the Bachelor Degree in *Chemical Engineering*, University of Calabria.

2018-2022 Lecturer of the course in *Solid Mechanics* for the Bachelor Degree in *Food Engineering*, University of Calabria.

2015-2017 Lecturer of the course in *Solid Mechanics* for the Bachelor Degree in *Technologies for the Conservation and Restoration of Cultural Heritage*, University of Calabria.

2015 Lecturer of the course in *FEM analysis of composite structures*, Post-graduate course at University of Naples 2.

2014 Lecturer of the course in *Dynamics of Structures* for the Master Degree in *Civil Engineering*, University of Calabria.

2011-2013 Lecturer of the course in *Computational Mechanics of Structures* for the Master Degree in *Civil Engineering*, University of Calabria.

2011-2013 Tutor of the course in *Theory of Structures* for the Master Degree in *Civil Engineering*, University of Calabria.

2007-2010 Lecturer of the course in *Solid Mechanics* for the Bachelor Degree in *Chemical Engineering*, University of Calabria.

2004-2010 Tutor of the course in *Solid Mechanics 2* for the Bachelor Degree in *Civil Engineering*, University of Calabria.

2003-2010 Tutor of the course in *Inelastic Analysis of Structures* for the Master Degree in *Civil Engineering*, University of Calabria.

2002-2006 Tutor of the course in *Theory of Structures* for the Master Degree in *Civil Engineering*, University of Calabria.

2007 Lecturer of the seminar in *Computational Modeling of Inelastic structures*, University of Basilicata.

2007 Lecturer of the seminar in *Damage Models for the FEM analysis of masonry*, University of Basilicata.

2004 Lecturer of the course in *Computational methods with FEM* for the Master Degree in *Civil Engineering*, University of Basilicata.

2003 Lecturer of the seminar in *FEM formulations for Structural Analysis*, University of Basilicata.

2002 Lecturer of the course in *Introduction to FEM*, Post-graduate course at University of Basilicata.

TEACHING BOOKS

- Bilotta A., Introduction to Solid Mechanics, https://www.mathworks.com/academia/courseware/a-first-course-on-solid-mechanics.html, web book base don the use of MATLAB (available in English and in Italian), 2020.

- Turco E., Casciaro R., Bilotta A., Porco F., Formica G., Scienza delle costruzioni, McGraw-Hill, Milano, 2004.

OTHER ACADEMIC ROLES

2013-present	Member of teaching staff of the Doctorate in “SCIENZE E TECNOLOGIE FISICHE, CHIMICHE E DEI MATERIALI” at University of Calabria, http://www.fis.unical.it/news.php?nid=1217#.XaXojC-B1yp.
2021-2026	Member of the paritetical commission teachers students of DIMES, University of Calabria.
2018-2020	Member of the Department Board (Department of Informatics, Modeling, Electronics and System Engineering - DIMES) at University of Calabria.
2009-2012	Secretary of Study Programs in Civil Engineering at University of Calabria.
2008-2013	Member of teaching staff of the Doctorate in “SCUOLA DI SCIENZA E TECNICA BERNARDINO TELESIO” at University of Calabria.
2006-2008	Member of teaching staff of the Doctorate in “SCUOLA DI DOTTORATO INTERNAZIONALE HARD SCIENCES BERNARDINO TELESIO” at University of Calabria.

SUPERVISOR ACTIVITY

2019	Sabina Mandaglio, Master Degree thesis keywords: *concrete materials, plasticity-based models.*
2010	Antonio L. Mendicino, PhD thesis keywords: *timber structures, moisture-stress, fracture.*
2008	Francesco Pasculli, Master Degree thesis keywords: *Hybrid FEM, Krylov methods, large structures.*
2007	Antonio L. Mendicino, Master Degree thesis keywords: *High continuity FEM, 3D elastic solids, ABAQUS.*
2005	Marialaura Malena, PhD thesis keywords: *brittle materials, mixed FEM.*
2003	Maria Grazia D’Aquila, Master Degree thesis keywords: *plasticity, holonomic path, Cam-clay.*
2001	Francesco Porco, PhD thesis keywords *Cosserat continuum, PUFEM.*
2001	Giampaolo Armentano, Master Degree thesis keywords: *plasticity, soils.*
1998	Francesco Porco, Master Degree thesis keywords: *Hybrid FEM.*

COMPUTER SOFTWARE SKILLS

- **C, C++**. Optimal knowledge of C and C++ programming languages with FEM software development experiences, also in collaborative environments, with linux, mac and windows OSs.
- **MATLAB**. Optimal knowledge.
- **Scientific computing libraries and environments**. Development of FEM software using gsl, petsc, slepc, mtl4 and mpi libraries. Beowulf cluster experiences. Experiences in using Mathematica, Maple and Sage.

- **FEM tools**. Experience of ABAQUS, Adina, Midas and gmsh softwares.
- **Other programming languages**. Knowledge and experience of html, php, RDBS systems (MySql), FORTRAN.

RELEVANT EXPERIENCES IN THE IMPLEMENTATION OF COMPILED SOFTWARE TOOLS FOR THE ANALYSIS OF STRUCTURES

2016-2019	Beam software. C++ code for the two-level FEM analysis of framed structures [17]. Nonlinear behaviour of the material modelled at the cross-section level with the possibility of describing generic composite materials. Code builded on the Gmsh, PETSc and SLEPc C libraries and open to the two-level analysis of other structural models.
2014-2015	TEAM (Tetrahedral Elements Analysis of Masonries) software. C++ code for the FEM analysis of generic 3D constructions constituted by chaotic masonries. Code builded on the Gmsh, PETSc and SLEPc C libraries and interfaced with laser-scanner output for the geometrical description of constructions (MADAR project).
2004-2005	MPI C++ code for intensive computations on Beowulf cluster [39, 38].
2001	Implementation of C and C++ libraries for the application of computational mechanics in structural design in civil and industrial fields (MECOM project).
1996-2000	KASP (Koiter Analysis of Slender Panels) software. C++ code for the FEM analysis of extruded composite panels modelled by using High Continuity finite elements for geometry and Koiter asymptotic approach to describe the structural nonlinear response. Prototipe software with Windows graphical user interface (APRICOS project).
1995-1996	Experience as programmer in Newsoft, engineering software house founded in 1979.
1994-1995	C++ coded software with Windows 3.1 graphical user interface for the FEM analysis of soils modelled as 2D domains with Cam-clay elasto-plastic response (Master Thesis “Analisi di continui elasto-plastici in condizione di deformazione piana in campo geotecnico”).

ARTICLES

---

[1] Bilotta A., Turco E., Estimating the mass density distribution from ultrasonic dispersion curves, Mathematics and Mechanics of Solids (2026). doi: 10.1177/10812865261441535

[2] Nitti A., Magisano D., Bilotta A., Leonetti L., Garcea G., De Tullio M. D., A fast isogeometric solver for electrophysiology based on weak spatial form with minimal patch-wise quadrature, Computer Methods in Applied Mechanics and Engineering, 449 (2026). doi: 10.1016/j.cma.2025.118548

[3] Turco E., Bilotta A., Inverse scattering transform: an overview and the Toda's chain as paradigm for discrete systems, Continuum Mech. Thermodyn. 37 (69) (2025). doi: 10.1007/s00161-025-01400-0

[4] Bertani G., Bilotta A., D'Altri A. M., de Miranda S., Liguori F. S., Madeo A., A hybrid stress finite element for the efficient nonlinear analysis of masonry walls based on a multi-failure strength domain, Finite Elements in Analysis and Design 244 (2025) 104310. doi: doi.org/10.1016/j.finel.2024.104310

[5] Bilotta A., Turco E., Constitutive modeling of heterogeneous materials by interpretable neural networks: A review, Networks and Heterogeneous Media 20 (1) (2025) 232-253. doi: 10.3934/nhm.2025012

[6] Turco E., Bilotta A., Designing multistable metamaterials inspired by Kresling tubes, Mechanics Research Communications 142, (2024) 104348. doi: 10.1016/j.mechrescom.2024.104348

[7] Bilotta A., Morassi A., Turco E., Damage identification for steel-concrete composite beams through convolutional neural networks. Journal of Vibration and Control, 30 (3-4) (2024) 876-889. doi:10.1177/10775463231152926

[8] Liguori F., Corrado A., Bilotta A., Madeo A., A layer-wise plasticity-based approach for the analysis of reinforced concrete shell structures using a mixed finite element, Engineering Structures, 285 (2023) 0-0. doi:10.1016/j.engstruct.2023.116045

[9] Bilotta A., Morassi A., Turco E., Simple convolutional neural networks for the damage identification in composite steel-concrete beams, Lecture Notes in Civil Engineering, 2023, 433 LNCE, pp. 422-431.

[10] Bilotta A., Causin A., Solci E., Turco E., Automatic description of rubble masonry geometries by machine learning based approach, In: Bonetti E., Cavterra C., Natalini R., Solci M. (eds) "Mathematical Modeling in Cultural Heritage", (2023) INdAM Series. Springer, in press.

[11] Ruggieri S., Liguori F.S., Leggieri V., Bilotta A., Madeo A., Casolo S., Uva G., An archetype-based automated procedure to derive global-local seismic fragility of masonry building aggregates: META-FORMA-XL, International Journal of Disaster Risk Reduction, 95 (2023) doi: 10.1016/j.ijdrr.2023.103903

[12] Liguori S.F., Corrado A., Bilotta A., Madeo A., An efficient plasticity-based model for reinforced concrete flat shells by a 4-nodes mixed finite element, Materials Research Proceedings, 26 (2023) 245-250.

[13] Liguori S.F., Corrado A., Bilotta A., Madeo A., An efficient elastoplastic model for the analysis of reinforced concrete shells, Procedia Structural Integrity, 44 (2022) 544-549.

[14] Bilotta A., Causin A., Solci E., Turco E., Representative Volume Elements for the Analysis of Concrete Like Materials by Computational Homogenization, In: Bonetti E., Cavterra C., Natalini R., Solci M. (eds) "Mathematical Modeling in Cultural Heritage", (2021) INdAM Series, vol 41. Springer, Cham. https://doi.org/10.1007/978-3-030-58077-3_2.

[15] Bilotta A., A MATLAB-based symbolic approach for the quick developing of nonlinear solid mechanics finite elements, chapter from the edited volume "Finite Elements Methods and Their Applications", (2020) IntechOpen, https://doi.org/10.5772/intechopen.94869.

[16] Bilotta A., Morassi A., Rosset E., Turco E., Vessella S., Numerical size estimates of inclusions in Kirchhoff-Love elastic plates, International Journal of Solids and Structures, 168 (2019) 58-72. doi 10.1016/j.ijsolstr.2019.03.006

[17] Bilotta A., Garcea G., A two-level computational approach for the elasto-plastic analysis of framed structures with composite cross-sections, Composite Structures, 209 (2019) 192-205. doi 10.1016/j.compstruct.2018.10.056

[18] Bilotta A., Morassi A., Turco E., The use of quasi-isospectral operators for damage detection in rods, Meccanica, 53(1-2) (2018) 319-345. doi 10.1007/s11012-017-0728-8

[19] Tedesco F., Bilotta A., Turco E., Multiscale 3D mixed FEM analysis of historical masonry constructions, European Journal of Environmental and Ingegneria civile, 21(7-8) (2017) 772-797. doi 10.1080/19648189.2015.1134676

[20] Bilotta A., Turco E., Elastoplastic analysis of pressure-sensitive materials by an effective three-dimensional mixed finite element, ZAMM Zeitschrift fur Angewandte Mathematik und Mechanik, 97(4) (2017) 382-396. doi 10.1002/zamm.201600051

[21] Bilotta A., Turco E., Numerical sensitivity analysis of corrosion detection, Mathematics and Mechanics of Solids, 22(1) (2017) 72-88. doi 10.1177/1081286514560093

[22] Bilotta A., Garcea G., Leonetti L., A composite mixed finite element model for the elasto-plastic analysis of 3D structural problems, Finite Elements in Analysis and Design, 113 (2016) 43-53.

[23] Bilotta A., Morassi A., Turco E., Reconstructing blockages in a symmetric duct via quasi-isospectral horn operators, Journal of Sound and Vibration, 113 (2016) 149-172. doi 10.1016/j.jsv.2015.12.038.

[24] Garcea G., Goncalves R., Bilotta A., Manta D., Bebiano R., Leonetti L., Magisano D., Camotim D., Deformation modes of thin-walled members: A comparison between the method of Generalized Eigenvectors and Generalized Beam Theory, Thin-Walled Structures, 100 (2016) 192-212. doi 10.1016/j.tws.2015.11.013

[25] Garcea G., Bilotta A., Leonetti L., An efficient algorithm for shakedown analysis based on equality constrained sequential quadratic programming, In: “Direct Methods for Limit and Shakedown Analysis of Structures: Advanced Computational Algorithms and Material Modelling”, Springer International Publishing, (2015)

[26] Genoese A., Genoese A., Bilotta A., Garcea G., Buckling analysis through a generalized beam model including section distortions, Thin-Walled Structures, 85 (2014) 125-141. doi 10.1016/j.tws.2014.08.012

[27] Genoese A., Genoese A., Bilotta A., Garcea G., A geometrically exact beam model with non-uniform warping coherently derived from the Saint Venant rod, Engineering Structures, 68 (2014) 33-46. doi: 10.1016/j.engstruct.2014.02.024

[28] Genoese A., Genoese A., Bilotta A., Garcea G., A composite beam model including variable warping effects derived from a generalized Saint Venant solution, Composite Structures, 110 (1) (2014) 140 - 151. doi: 10.1016/j.compstruct.2013.11.020

[29] Genoese A., Genoese A., Bilotta A., Garcea G., A generalized model for heterogeneous and anisotropic beams including section distortions, Thin-Walled Structures, 74 (2014) 85-103. doi 10.1016/j.tws.2013.09.019

[30] Garcea G., Bilotta A., Madeo A., Casciaro R., Direct evaluation of the post-buckling behavior of slender structures through a numerical asymptotic formulation, In: “Direct Methods for Limit States in Structures and Materials”, Springer International Publishing, (2013)

[31] Genoese A., Genoese A., Bilotta A., Garcea G., A mixed beam model with non-uniform warpings derived from the Saint Venant rod, Computers and Structures, 121 (2013) 87-98.

[32] Bilotta A., Leonetti L., Garcea G., An algorithm for incremental elastoplastic analysis using equality constrained sequential quadratic programming, Computers and Structures, 102-103 (2012) 97-107.

[33] Bilotta A., Leonetti L., Garcea G., Three field finite elements for the elastoplastic analysis of 2D continua, Finite Elements in Analysis & Design, 47 (2011) 1119-1130.

[34] Bilotta A., Formica G., Turco E., Performance of a high-continuity finite element in three-dimensional elasticity, International Journal for Numerical Methods in Biomedical Engineering, 26 (2010) 1155-1175 .

[35] Bilotta A., Turco E., A numerical study on the solution of the Cauchy problem in elasticity, International Journal of Solids and Structures, 46 (2009) 4451-4477.

[36] Alessandrini G., Bilotta A., Morassi A., Rosset E., Turco E., Computing volume bounds of inclusions by EIT measurements, Journal of Scientific Computing, 33 (2007) 293-312.

[37] Bilotta A., Casciaro R., A high performance element for the analysis of 2D elastoplastic continua, Comput. Methods Appl. Mech. Engrg., 196 (2007) 818-828.

[38] Alessandrini G., Bilotta A., Formica G., Morassi A., Rosset E., Turco E., Evaluating the volume of hidden inclusions in an elastic body, Journal of Computational and Applied Mathematics, 198 (2007) 288-306.

[39] Alessandrini G, Bilotta A., Formica G., Morassi A., Rosset E., Turco E., Numerical size estimates of inclusions in elastic bodies, Inverse Problems, 21 (2005) 133-151.

[40] Fortino S., Bilotta A., Evaluation of the amount of crack growth in 2D LEFM problems, Engineering Fracture Mechanics, 71/9-10 (2004) 1403-1419.

[41] Bilotta A., Casciaro R., Assumed stress formulation of high order quadrilateral elements with an improved in-plane bending behaviour, Comput. Methods Appl. Mech. Engrg., 191/15-16 (2002) 1523-1540.

[42] Salerno G., Bilotta A., Porco F., A finite element with micro-scale effects for the linear analysis of brick masonry structures, Comput. Methods Appl. Mech. Engrg., 190/34 (2001) 4365-4378.

INTERNATIONAL CONFERENCE CONTRIBUTIONS

[43] A. Bilotta, E. Turco, Dispersion based identification of mass density distributions via inverse parametrized eigenvalue solutions, IPMS 2026, Twelfth International Conference Inverse Problems: Modeling and Simulation, Mellieha, Malta, May 31 - June 5, 2026.

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